



Chapter 1

Introduction to Computer Networks

What You Will Learn

- ☐ Introduction
- ☐ Network Hardware
- ☐ Network Software
- ☐ Reference Models – OSI & TCP/IP

What is a Computer Network?

- ❑ A collection of autonomous computers interconnected by a single technology.
- ❑ Provides communication that is
 - ◎Reliable- takes care of faults
 - ◎Fair - integrity of message communicate
 - ◎Efficient- works at all times
 - ◎From one application to another- client to server
- ❑ Automatically detects and corrects
 - ◎Data corruption- error detection and correction codes
 - ◎Data loss- Data sent but not received is retransmitted
 - ◎Duplication- multiple copies at receiver will be discarded
 - ◎Out-of-order delivery- will be sequenced in some order
- ❑ Automatically finds optimal path from source to destination

Internetworking

A collection of autonomous computers interconnected by a single technology is called computer networks.

Two computers are said to be interconnected if they are able to exchange information via copper wire/ fibreoptics/ microwaves/ infrared/ communication satellites.

Networks can be of any size/ shape / form.

Inter-networks means a network of networks.

Distributed system- a collection of independent computers appears to its users as a single coherent system.

Network examples

- ❑ Telephone network
- ❑ Satellite network
- ❑ TV programs- cable network
- ❑ Internet – wireless network
 - © ftp - file transfer protocol
 - © Mail - mail transfer protocol
 - © Chat- TCP/UDP
 - © ...

Uses of Computer Networks

❑ Business Applications

- ◎ online buying- e-commerce
- ◎ sharing physical resource like printer on network
- ◎VPN - Virtual Private Networks- joining two distant network into an extended network
- ◎VoIP- Voice over IP -saved on telephone bills
- ◎Desktop sharing- use shared whiteboard

❑ Home Applications

- ◎ mail, chat, newspapers ,books, research papers available online
- ◎Peer to peer network - BitTorrent no central database.
- ◎Instant messaging- Twitter, internet Radio and Video(YouTube), Social networking-Facebook, linkedIn wikipedia

Uses of Computer Networks

❑ Mobile Users

- ◎ wireless communication with laptops, PDA, mobile, or in plane is easier.
- ◎ Taxi booking in metropolis - Ola, Uber..

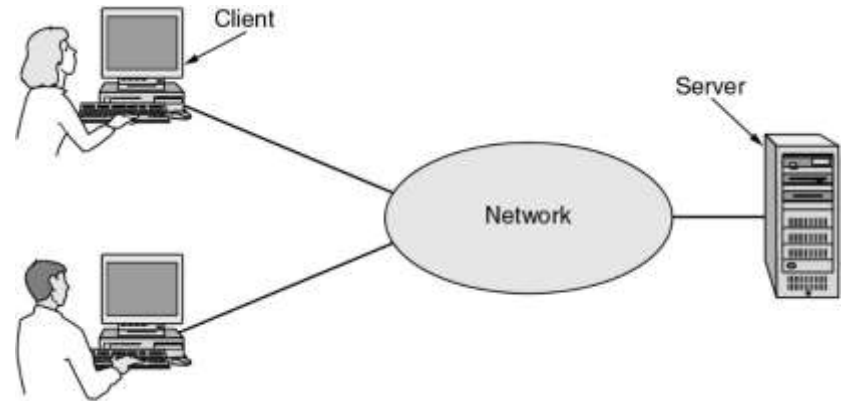
❑ Social Issues

- ◎ Network Neutrality - practice of providing similar service no matter high or low profile user or depending on the content or source.
- ◎ Digital Millennium Copyright Act - US govt issue DMCA takedown notices to peer-to peer networks carrying pirated contents(music, video)
- ◎ Profiling users by private sector, and collecting data about user via cookies helps them target specific Ads

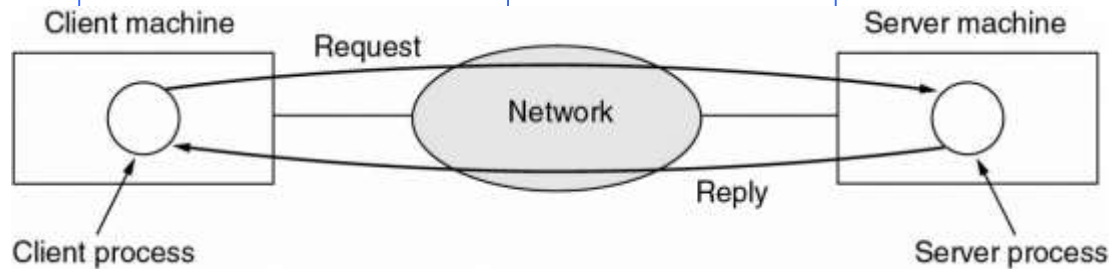
Business Applications of Networks

❑ A network with two clients and one server.

- ◎ Check bank account
- ◎ Pay bills
- ◎ Reserve ticket



❑ The client-server model involves requests and replies (web application).



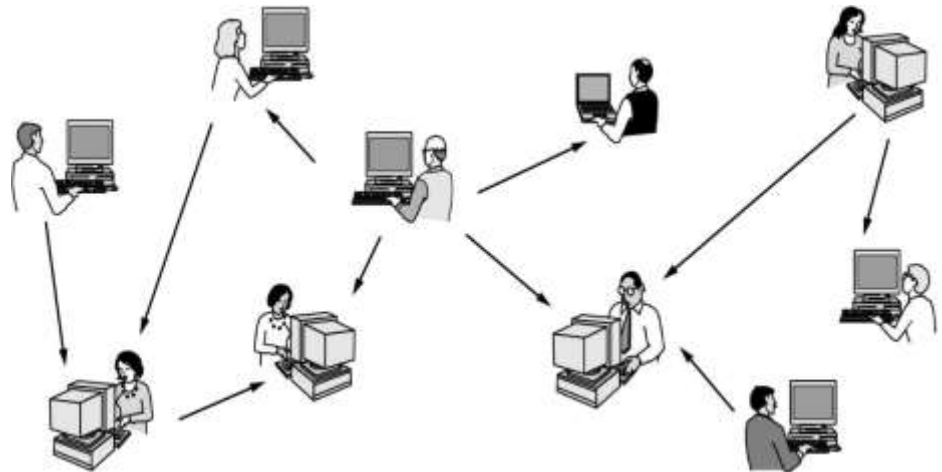
Home Network Applications

- ❑ Access to remote information
 - ◎ Leaning online, downloading
- ❑ Person-to-person communication
 - ◎ chat, phone
- ❑ Interactive entertainment
 - ◎ games, movies, ...
- ❑ Electronic commerce

Home Network Applications (2)

❑ Peer-to-peer (P2P)

©Kazaa, Emule,



❑ E-commerce

Full name	Example
Business-to-consumer	Ordering books on-line
Business-to-business	Car manufacturer ordering tires from supplier
Government-to-consumer	Government distributing tax forms electronically
Consumer-to-consumer	Auctioning second-hand products on-line
Peer-to-peer	File sharing

Mobile Network Users

Combinations of **wireless** networks and mobile computing.

❑ Desktop Computers are non movable computers like in our college

Wireless	Mobile	Applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	Yes	Portable office; PDA for store inventory

❑ A Notebook/Laptop can be moved around and wired

❑ Networks using access points (without wires) third case but fixed

❑ PDA –Personal Digital Assistants- like tabs

Mobile Network Users

- ❑ Text messaging or texting- SMS – Short Message service – Telecoms gain revenues by charging SMS heavily (outside US)
- ❑ Smart Phones- can provide fast data(using 3G or 4G Technology) to browse, to book tickets, to check email, to watch movies or listen to music as well as phone calling
- ❑ Global Positioning System – to trace the position of the mobile on the globe- for guiding while travelling, and check nearby hotels or bookstore
- ❑ M-commerce- purchase and sell with mobile apps like Olx, banking on mobile, pay bills etc.
- ❑ NFC- Near Field Communication – by having the RFID enabled mobile come near a POS reader terminal for payment , can check the price of the products and compare with others.
- ❑ Sensor Networks – predecessor of IOT- car may have many sensors- to find location , speed , vibration and fuel efficiency from its onboard diagnostic system and save it in a remote database and later help you find potholes, plan trips around congested roads and tell if you are a ‘gas guzzler’
- ❑ Wearable Computers- PDA’s , smart watches , tabs, pace makers and insulin pumps.

Social Issues

- ❑ Discussions about
 - ◎ politics,
 - ◎ religion,
- ❑ Hack and robbery.
- ❑ Electronic Junk Mail (spam)-few have collected e-mail addresses of many users and are sending marketing messages to them.
- ❑ Botnet- set of compromised machines sending spam messages or steal your bank account passwords.
- ❑ Phishing-messages masquerade as originating from a trustworthy party like your bank to try to trick you into revealing sensitive information



Network Hardware

Keywords: Point to Point, Broadcast, LAN, WAN, Wired vs Wireless

What A Network Includes

- ❑ Transmission hardware
- ❑ Special-purpose hardware devices
 - ⊙ interconnect transmission media
 - ⊙ control transmission
 - ⊙ run protocol software
- ❑ Protocol software
 - ⊙ encodes and formats data
 - ⊙ detects and corrects problems

Network Hardware

❑ Transmission technology (2 types)

- ◎ Broadcast links- every terminal connected to many other terminals and can send and receive packets. unicast, multicast and broadcasting
- ◎ Point-to-point links- packets from one sender to one receiver is called unicasting

❑ Scale

- ◎ Local Area Networks (LAN)
- ◎ Metropolitan Area Networks (MAN)
- ◎ Wide Area Networks (WAN)
- ◎ Wireless Networks
- ◎ Home Networks
- ◎ Internetworks

❑ Media

- ◎ Wire line
- ◎ Wireless

Broadcast Networks



❑ There are **A single communication link** for all systems in network = **Broadcasting**

© TV programs: Doordarshan could be accessed with antenna...

❑ Messages (**Packets**) contain destination address

❑ **Multicasting**: A subset of systems can get the message

❑ Usually used in small networks like LANs

Point-to-point

- ❑ Individual connections between pairs of machines.
- ❑ There are many **paths** from one machine to another
 - ◎ Need efficient **routing algorithms**
- ❑ Usually used in large scale networks like WAN



So ...

❑ Packets

- © Messages - the "chunk" of data transmitted from one machine to the next.

❑ Addressing

- © One to one: Packet contains specific target address.
- © Broadcasting: All machines on the network receive and process the packet.
- © Multicasting: A subset of machines receive and process the packet

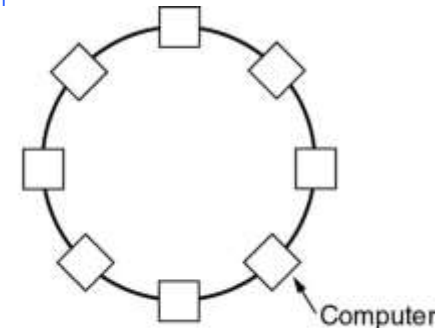
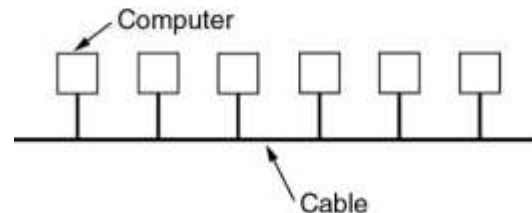
Classification by scale

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Local Area Networks (1)

- ❑ Privately owned. Can be up to several kilometers long;
Ex. in a building
- ❑ Separated by their:

- 1 Size: Restricted so worst case transmission time can be contained.
- 2 Transmission technology: Single channel with multiple machines connected to it. Run at speeds of 10, 100, or more Mbps.
- 3 Topology: two popular broadcast networks:
 - Bus
 - Ring



Local Area Networks (2)

☐ Topology ...

© Bus

➤ Ethernet (IEEE 802.3):

- Bus based broadcast network with decentralized control at 10 or 100 Mbps.

© Ring

➤ Token Ring (IEEE 802.5):

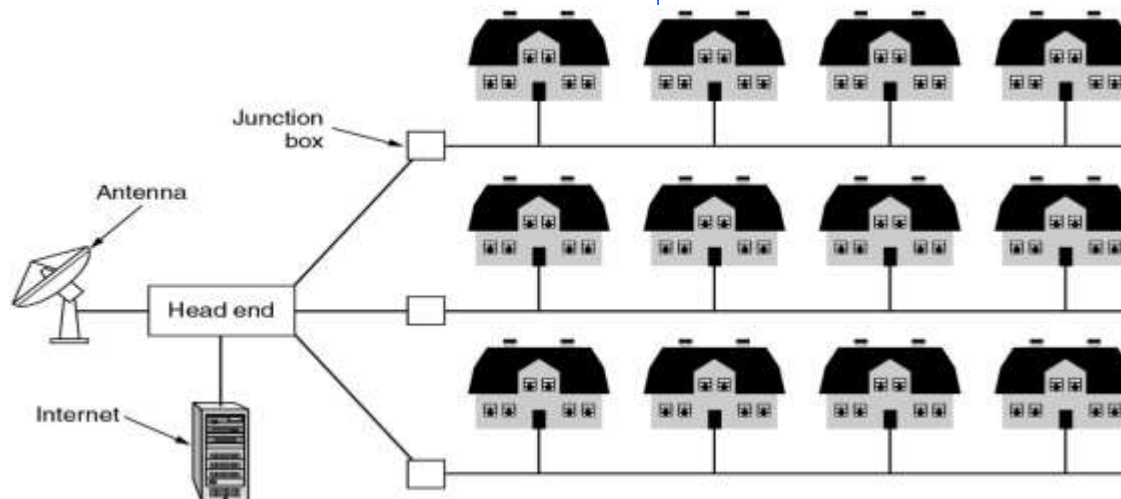
- Ring based broadcast network with token arbitration at 4 or 16 Mbps.

☒ Low delay. High reliability.

☐ Requires collision arbitration

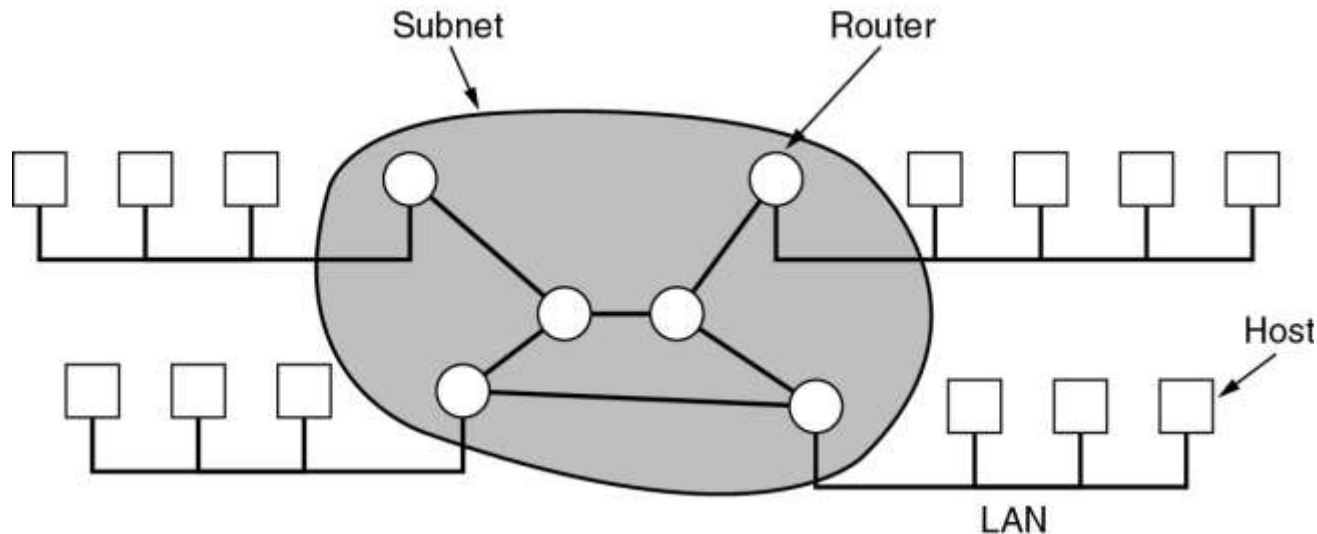
Metropolitan Area Networks

- ❑ Larger version of LAN ("city" wide).
 - ◎ Public or private / data or voice.
 - ◎ Broadcast - no switches.
 - ◎ Can be distinguished from LANs based on wiring mechanism.
 - **DQDB** (Distributed Queue Dual Bus), IEEE 802.6
- ❑ Ex. A metropolitan area network based on cable TV



Wide Area Networks (1)

- ❑ Networks spanning large distances.
- ❑ Ex. Relation between hosts on LANs and the subnet.



- ❑ **Hosts** or End Systems:
 - © Machines running user applications.

Wide Area Networks (2)

❑ (Communication) Subnet:

- ◎ Connections between hosts - transmission lines + switches.
- ◎ A "locality" understanding each other's addresses.

❑ Circuits (Channels, Trunks):

- ◎ Transmission lines move the bits.

❑ Packet switching nodes (Router, Intermediate systems):

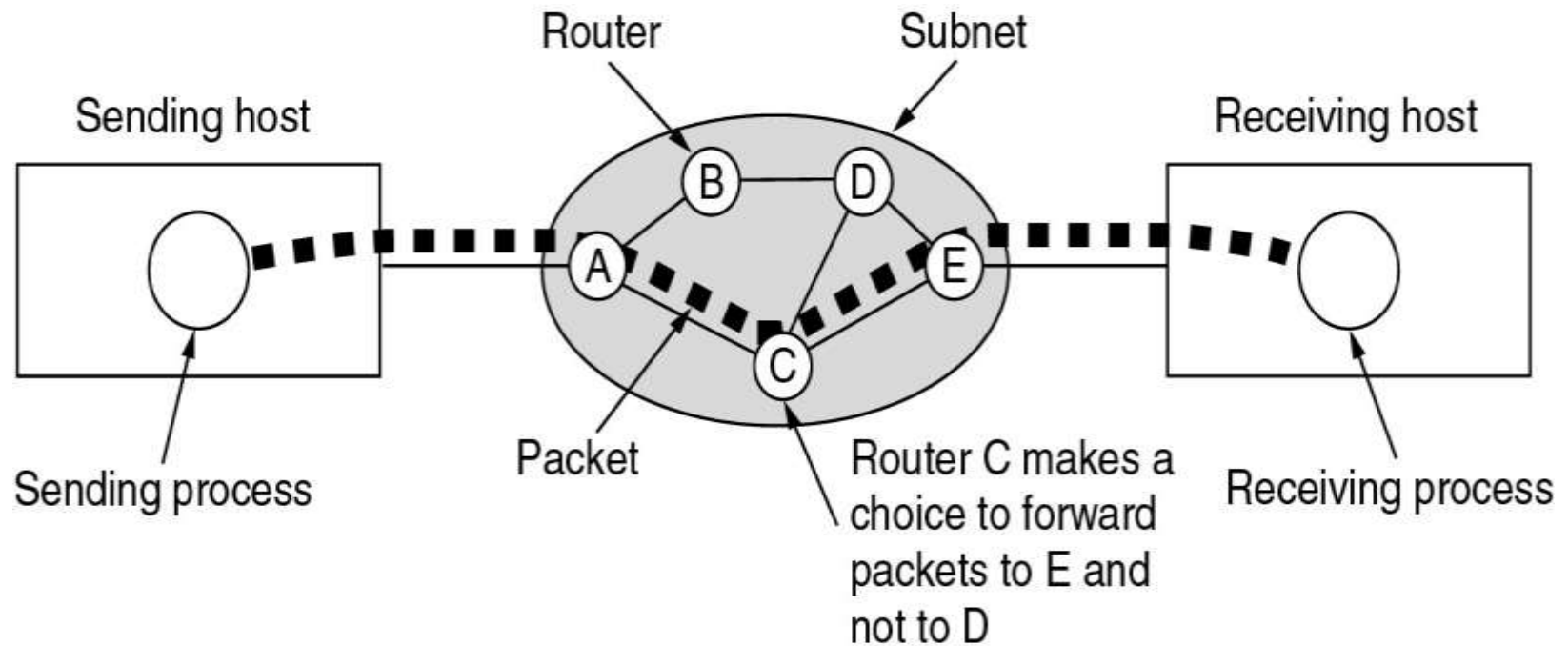
- ◎ Specialized computers moving data between several inputs to several outputs.

❑ Point-to-point/Store-and-forward/Packet-switched -

- ◎ Moving through a series of routers, packets are received at a router, stored there, then forwarded to the next router.

Wide Area Networks (3)

□ Ex. A stream of packets from sender to receiver.



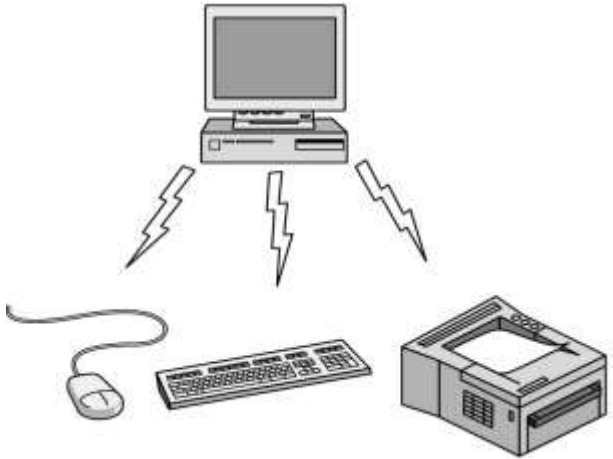
Wireless Networks (1)

- ❑ Used where computer is mobile or far away from wires.
 - ◎ Only 1 - 2 Mbps,
 - ◎ higher error rates,
 - ◎ interference

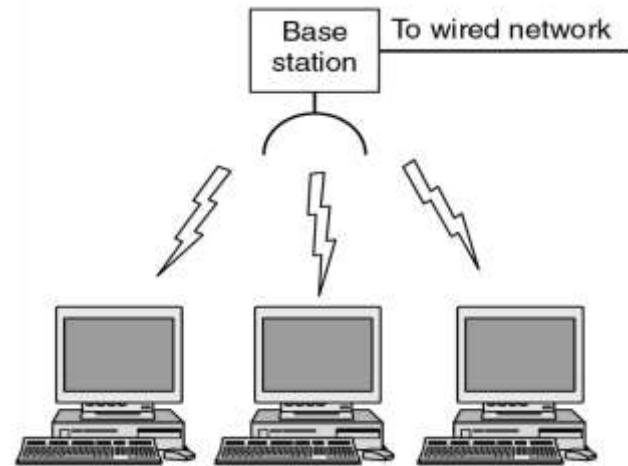
- ❑ Use
 - ◎ Sound
 - ◎ Light and mirrors
 - ◎ Infrared
 - ◎ RF
 - ◎ Microwave

Wireless Networks (2)

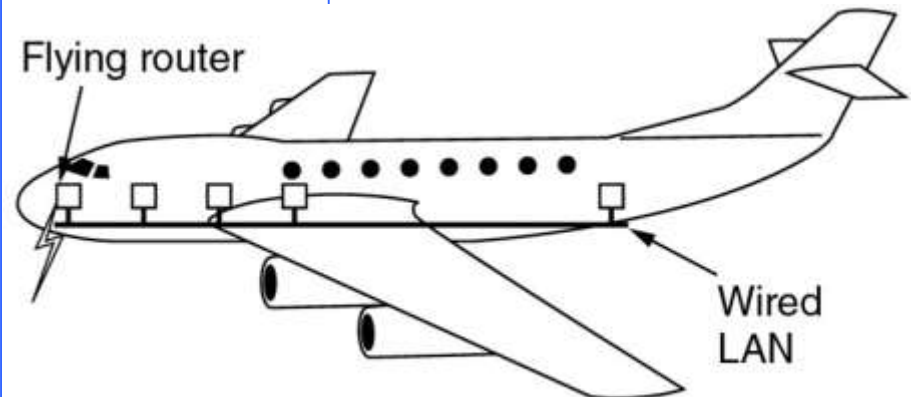
❑ Bluetooth configuration



❑ Wireless LAN



❑ Ex. A flying LAN





Network Software

Keywords: Layering, Protocol, interfaces,
Design issues of layers

Network Software (1)

- ❑ Primary networks more depend on hardware
- ❑ It talks about the philosophy of connecting together two entities.

- ❑ “Layering” is the key word
 - ©Protocol Hierarchies
 - ©Design Issues for the Layers
 - ©The Relationship of Services to Protocols

Network Software (2)

❑ Layers :

©The concept that network software is organized functionally into levels. A level on one host talks to the same level on another host (its peer).

❑ Protocol :

©The protocol is the convention or standard that a layer uses to talk to the other layer. An agreement or standard on the conversation.

Network Software (3)

□ Protocol Hierarchies

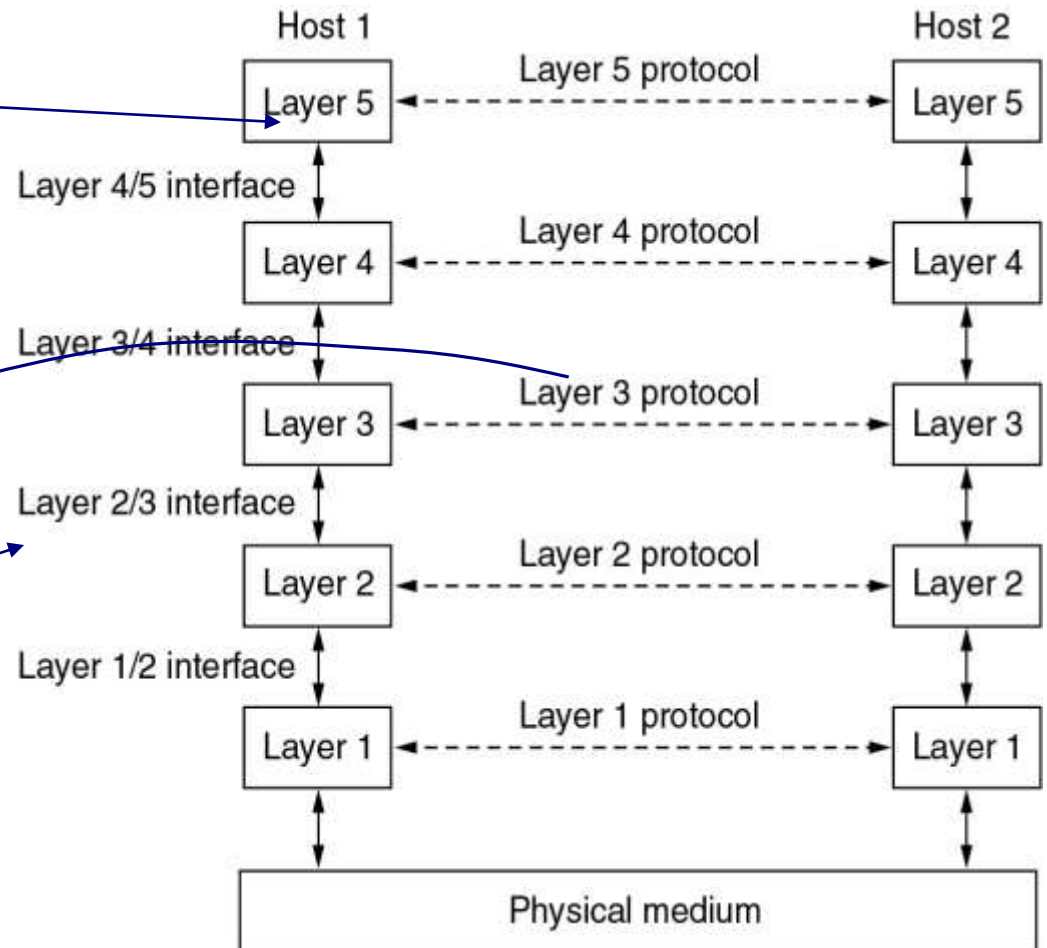
©Layers,

Important that each layer perform specific actions.

©protocols,

©Interfaces

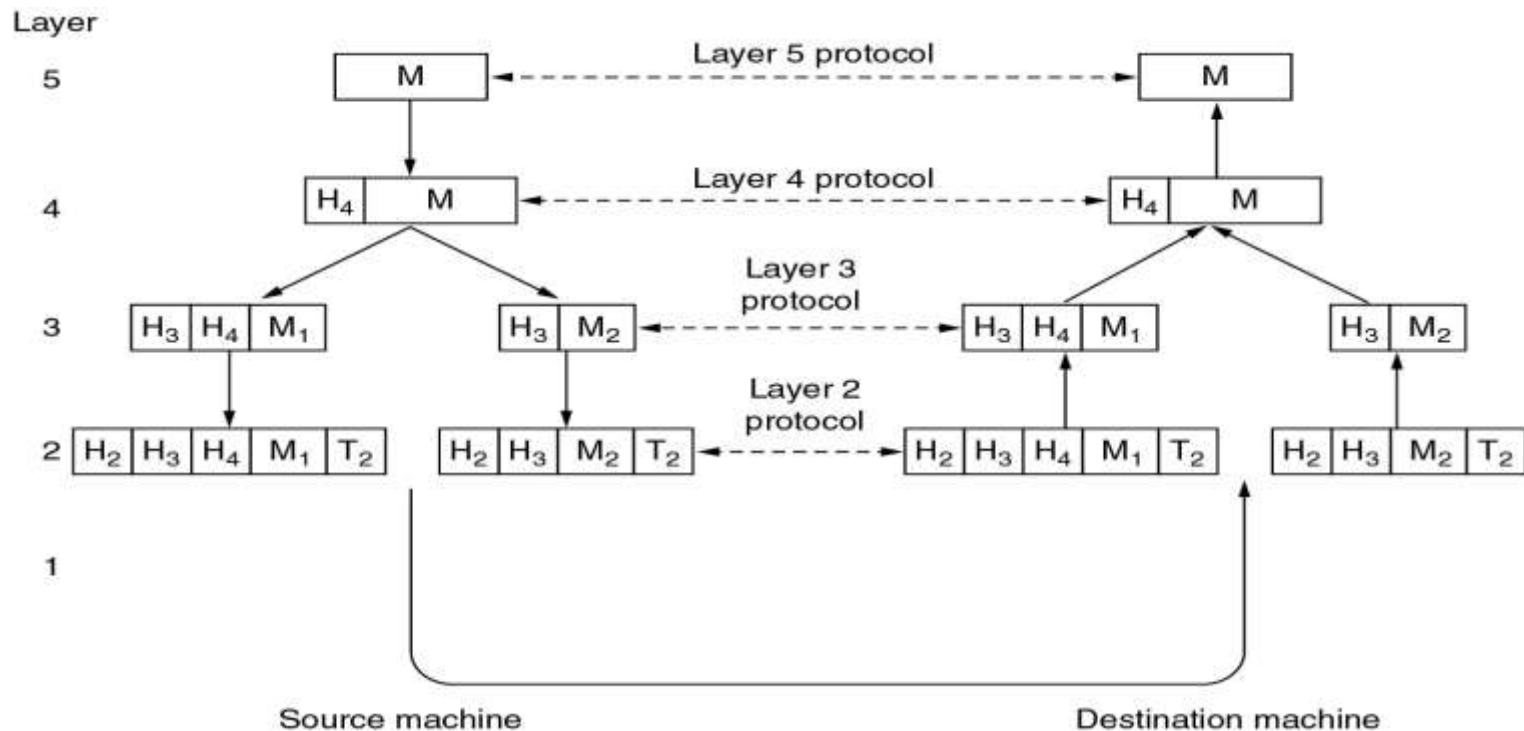
Defines the services That one layer offers another (either up or down.)



Network Software (5)

□ Ex. Protocol Hierarchies

◎ information flow supporting virtual communication in layer 5



Network Software (6)

❑ Physical Medium:

- ◎ Underneath the layers is the wire or fiber or whatever.

❑ Network architecture:

- ◎ A set of layers and protocols. It contains details on what happens in the layer and what the layers says to its peer.
- ◎ Functional interfaces and implementation details are not part of the spec, since that's not visible outside the machine.

❑ Protocol stack:

- ◎ A list of protocols used by a system, one protocol per layer.

❑ Information flow:

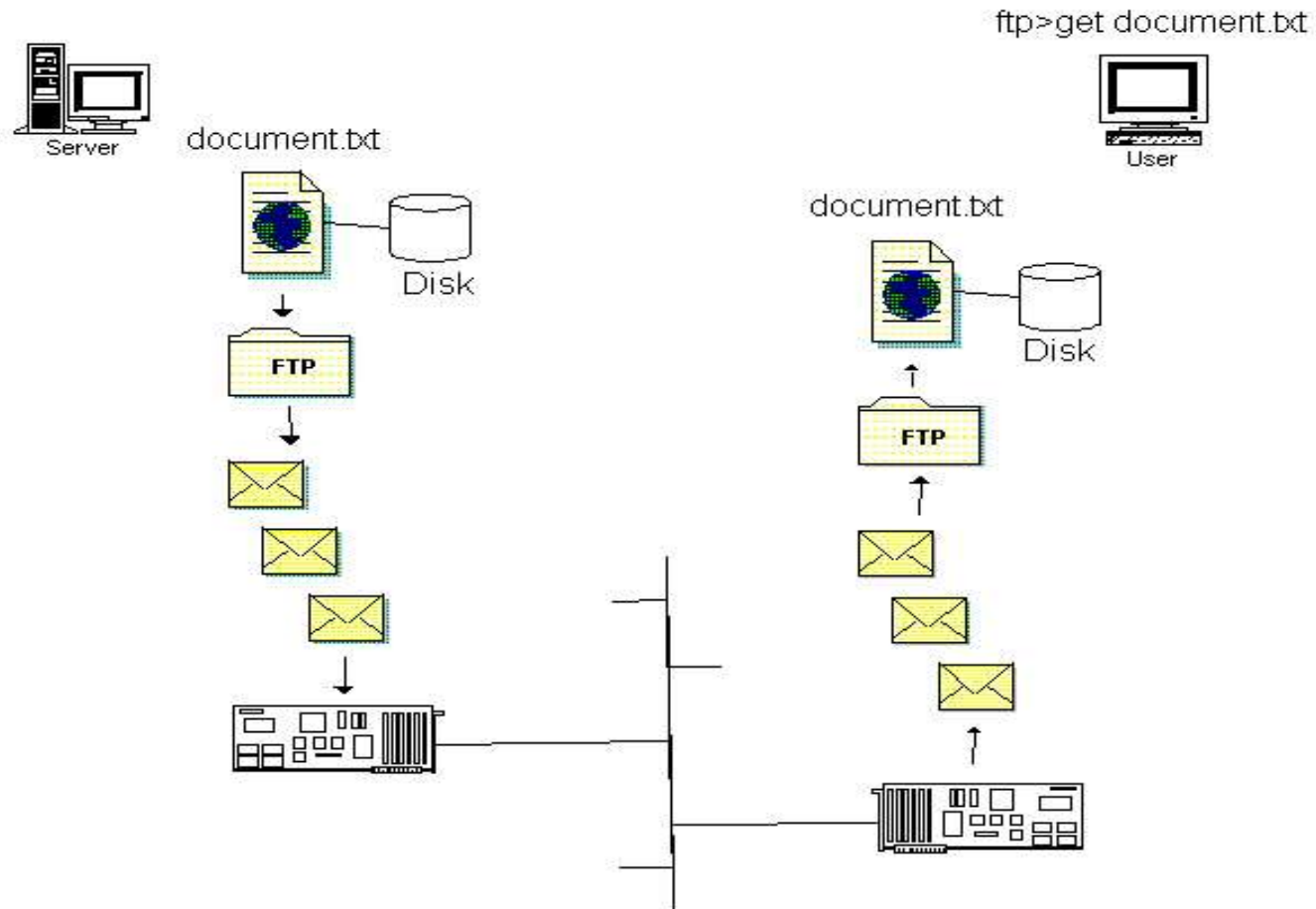
- ◎ "Send_to_peer" rather than "call_next_layer_down".

Design Issues for the Layers

Connection	Both Directions	Simultaneous
Simplex	No	No
Half duplex	Yes	No
Full duplex	Yes	Yes

- ☐ Addressing
- ☐ Error control. (garbled or missing.)
- ☐ Preservation of message ordering.
- ☐ Flow control.
- ☐ Breaking up messages into a smaller chunks (and reassembly.)
- ☐ Multiplexing messages on same connection.
- ☐ Routing - how to get from one host to another.

Layering





Reference Models

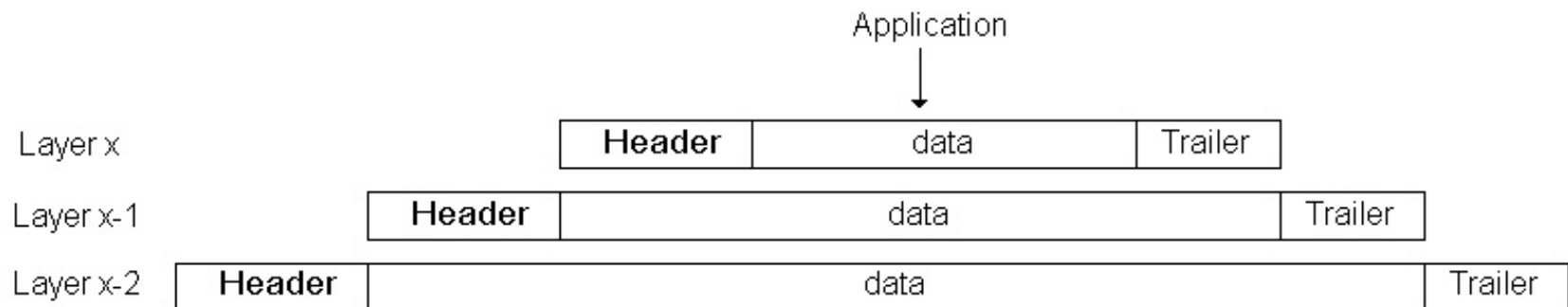
Keywords: OSI , TCP/IP

Reference Models

❑ Headers, Data, and Trailers



❑ Encapsulation



Reference Models (1)

- ❑ There are two competing models for how the software is layered. These are the **OSI** and the **TCP** models.

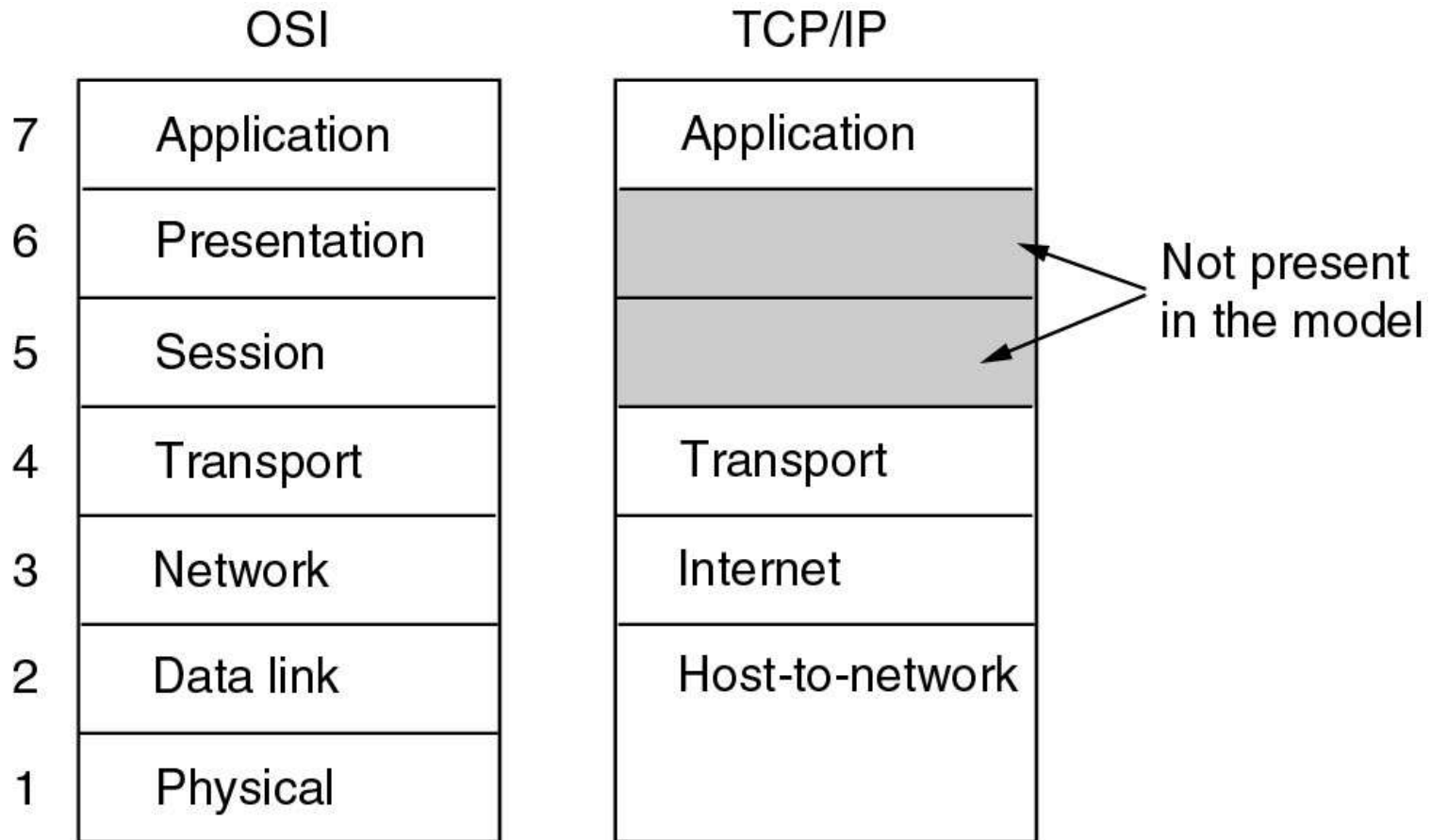
- ❑ **OSI (Open Systems Interconnection)**

- © Developed by ISO (International Standards Organization)
- © 7 layers

- ❑ **TCP (Transfer Control Protocol)**

- © Used in the Arpanet and in the **Internet**. Common mechanism that is surpassing the OSI Model.
- © 5 layers

Reference Models (2)



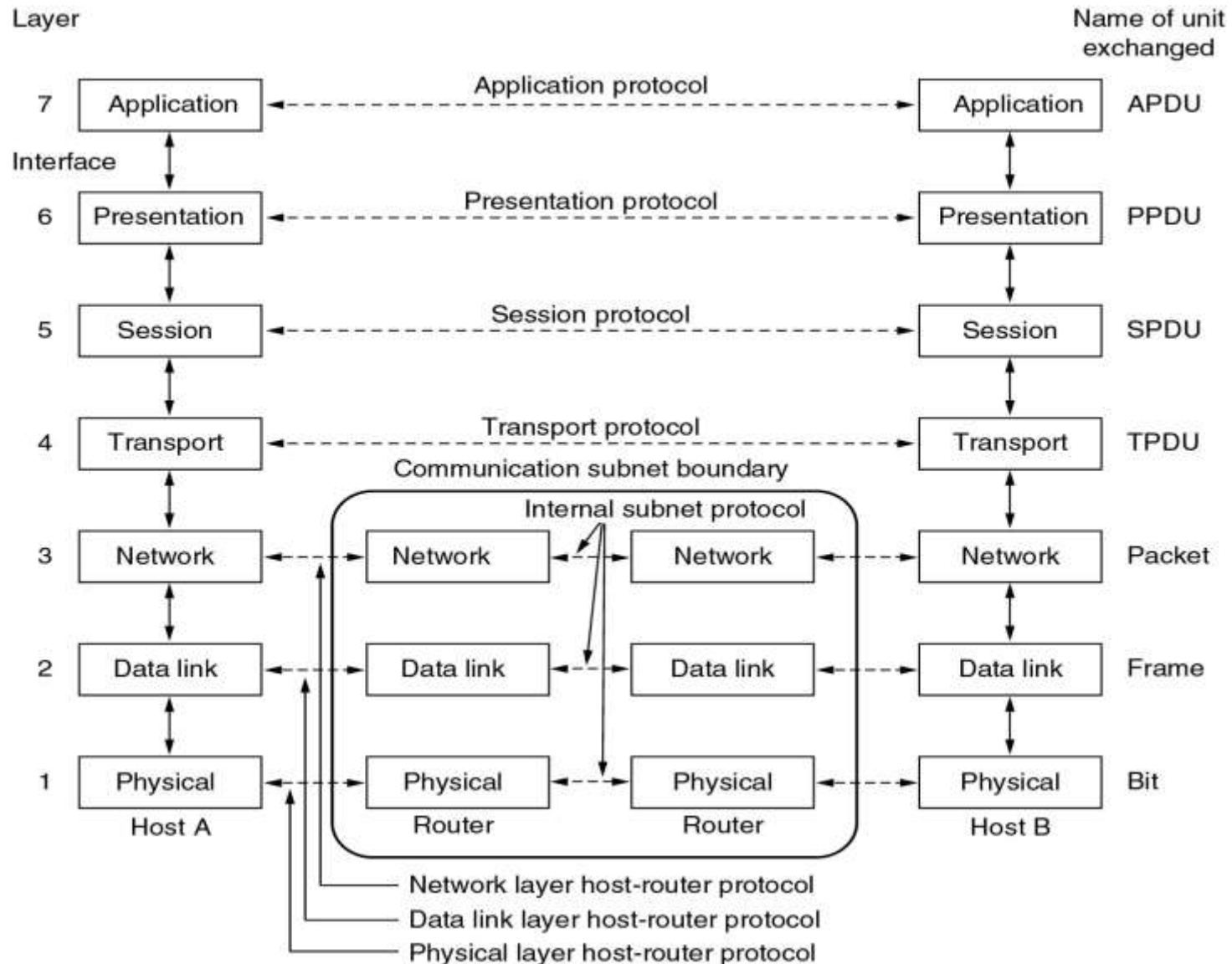
In the TCP/IP layer, the presentation and session layers are not present

OSI Reference Model (1)

□ Principles used to develop OSI Layering:

1. Need a layer for each different level of abstraction.
2. Each layer performs a well defined function.
3. Each layer should be standardizable.
4. Layer boundaries should minimize data flow across those boundaries.
5. The right number of layers - don't put too many functions together, but not too many layers either.

OSI Reference Model (2)



OSI Reference Model (3)

1 Physical Layer:

- © Purpose: Transmits raw **bits** across a medium.
- © Electrical: Concerns are voltage, timing, duplexing, connectors, etc.

2 Data Link Layer:

- © **Framing**: Breaks apart messages into **frames**. Reassembles frames into messages.
- © **Error handling**: solves damaged, lost, and duplicate frames.
- © **Flow control**: keeps a fast transmitter from flooding a slow receiver.
- © **Gaining Access**: if many hosts have usage of the medium, how is access arbitrated.

OSI Reference Model (4)

3 Network Layer:

- © **Routing**: What path is followed by packets from source to destination. Can be based on a static table, when the connection is created, or when each packet is sent.
- © **Congestion**: Controls the number packets in the subnet.
- © **Accounting**: Counts packets/bytes for billing purposes.

OSI Reference Model (5)

4 Transport Layer:

- © **Reliability**: Ensures that packets arrive at their destination. Reassembles out of order messages.
- © **Hides network**: Allows details of the network to be hidden from higher level layers.
- © **Service Decisions**: What type of service to provide; error-free point to point, datagram, etc.
- © **Mapping**: Determines which messages belong to which connections.
- © **Naming**: "Send to node ZZZ" must be translated into an internal address and route.
- © **Flow control**: keeps a fast transmitter from flooding a slow receiver.

OSI Reference Model (6)

5 Session Layer:

- © **Sessions**: Provides services that span a particular message. For instance, a login session could be logged.
- © **Synchronization**: Provide way to subdivide a long mechanism for reliability.

6 Presentation Layer:

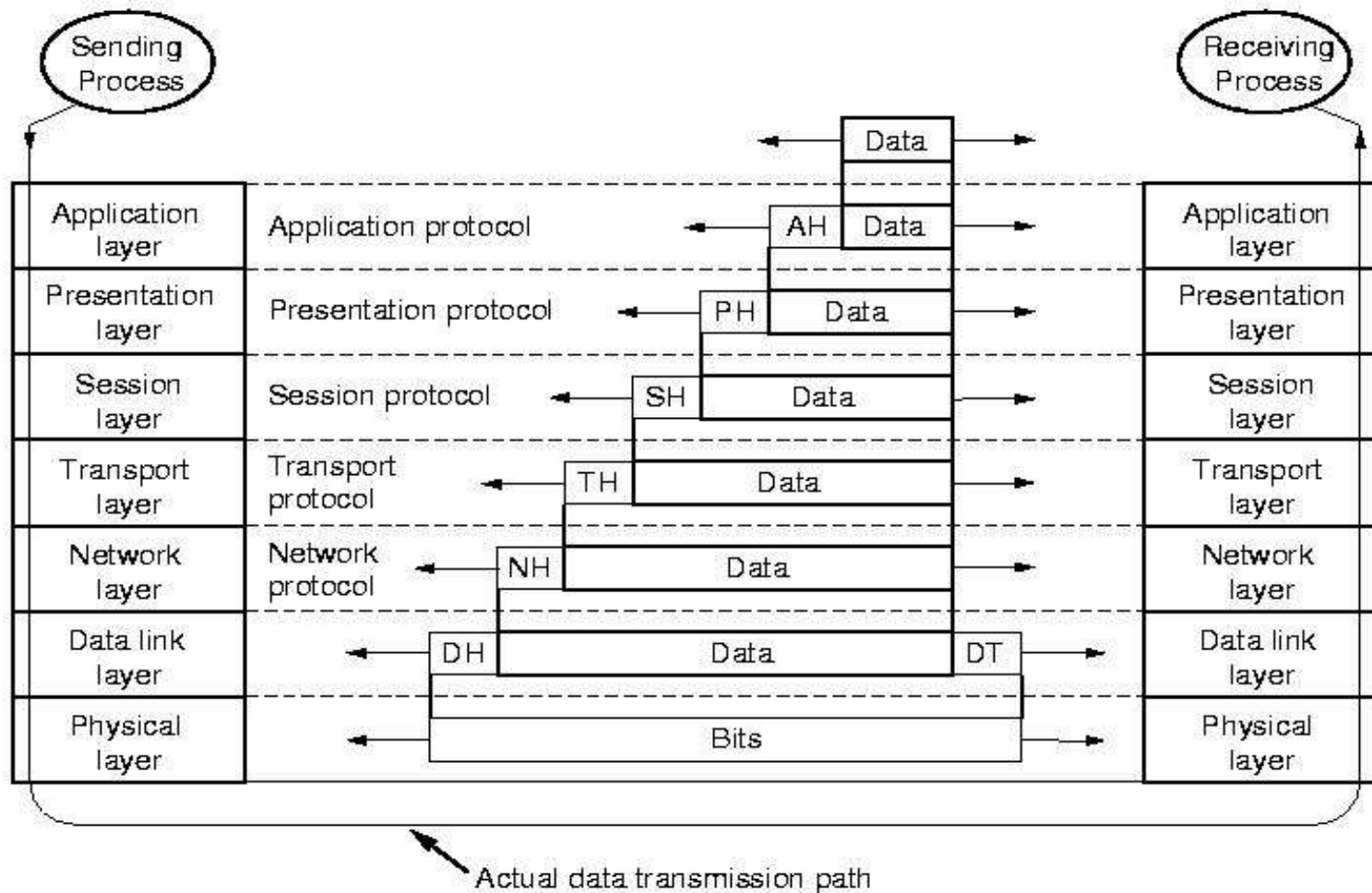
- © **Prettiness**: Syntax and semantics of information transmitted. Understands the nature of the data being transmitted. Converts ASCII/EBCDIC, big endian/little endian

7 Application Layer:

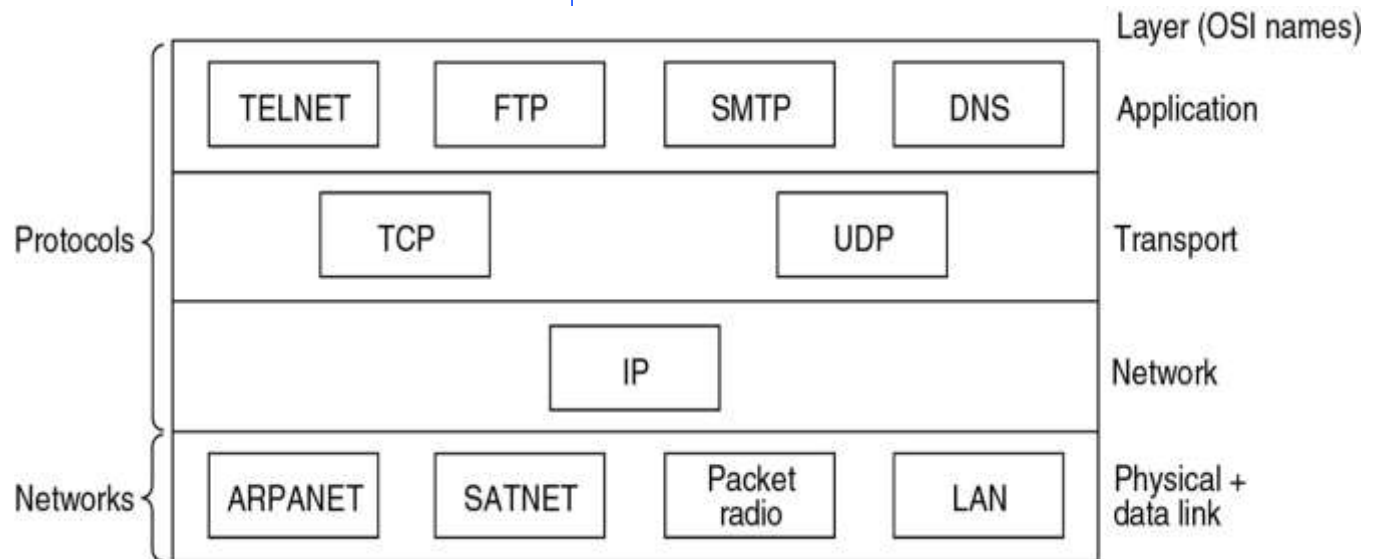
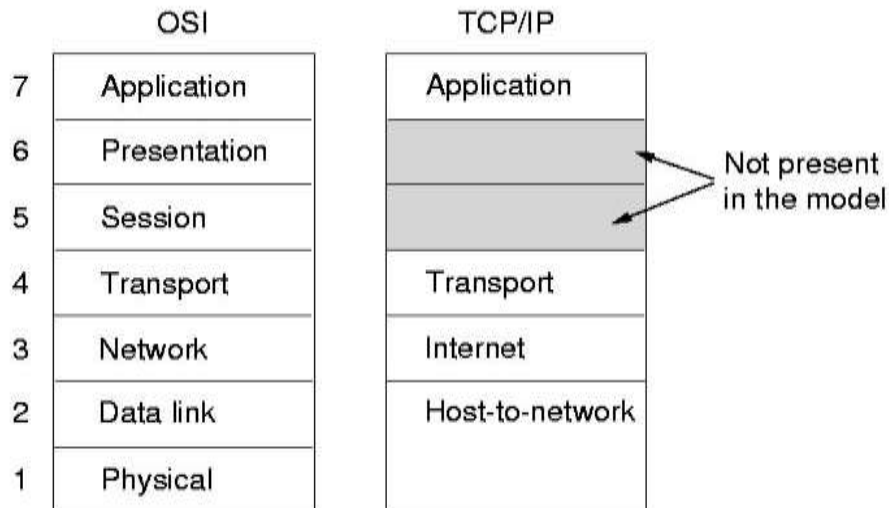
- © **Interfacing**: Terminal type translation.
- © **File transfer**: Programs able to understand directory structures and naming conventions and map them onto various systems.

OSI Reference Model (7)

□ Data Transmission in the OSI Model



TCP/IP Reference Model (1)



TCP/IP Reference Model (2)

2 Internet Layer

- ◎ **Connector**: Provides packet switched connectionless service.
- ◎ **Routing**: The IP (Internet Protocol) does delivery and congestion control.

3 Transport Layer

- ◎ **TCP (Transmission Control Protocol)**: provides a reliable connection oriented protocol that delivers a byte stream from one node to another. Guarantees delivery and provides flow control.
- ◎ **UDP (User Datagram Protocol)** provides an unreliable connection-less protocol for applications that provide their own.

TCP/IP Reference Model (3)

4 Application Layer

©Terminal Telnet

©File transfer FTP

©The Web HTTP

©Mail SMTP



End of Introduction

Next chapter: Physical Layer